Q1. Difference between Classification and Clustering

Ans:

Classification	Clustering
Supervised learning	Unsupervised learning
Uses labeled data	Uses unlabeled data
Predicts predefined classes	Finds hidden groups
Output is known categories	Output is unknown clusters
Example: Spam detection	Example: Customer grouping

Q2. Generate association rules from transaction data and calculate support and confidence

Ans:

Example Transaction Data

TID

Items

1 Milk, Bread

2 Milk, Butter

3 Bread, Butter

4 Milk, Bread, Butter

Rule: Milk → Bread

Support

Support = (Transactions containing Milk & Bread) / Total Transactions

= 2 / 4 = 0.5

Confidence

Confidence = (Transactions containing Milk & Bread) / (Transactions containing Milk)

= 2 / 3 = 0.66

Or

Q3. Apply Market Basket Analysis using a retail store example

Ans:

Example (Retail Store Data)

Customer 1: Rice, Dal

Customer 2: Rice, Oil

Customer 3: Dal, Oil

Customer 4: Rice, Dal, Oil

Association Rules

Rice → Dal

Dal → Oil

Rice → Oil

Insight

Customers buying rice often also buy dal

Dal buyers frequently purchase oil

Application

Place Rice, Dal, Oil close together

Offer combo packs (Rice + Dal)

Suggest “frequently bought together” items

Q4.Define the following terms

Ans:

1. Star Schema

A data warehouse structure with one central fact table connected to multiple dimension tables. Simple and fast for queries.

2. Snowflake Schema

An extension of star schema where dimension tables are further divided into sub-tables (normalized). More complex but saves space.

3. Fact Table

Stores numerical data (measures) like sales amount, quantity, etc., along with foreign keys to dimension tables.

4. Dimension Table

Stores descriptive data like product name, customer, time, location.

5. ETL Process

Process of Extracting data from sources, Transforming it into useful format, and Loading it into a data warehouse.

Q5 ,Q6, Q7 any Two

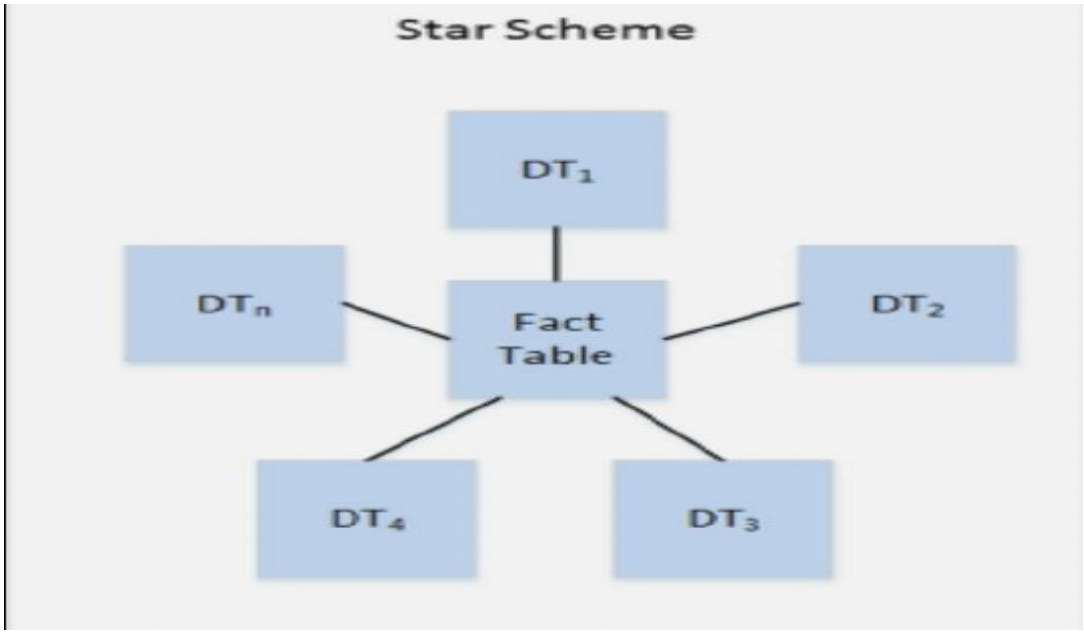
Q5. Explain the structure of Star Schema with diagram

Ans:

Structure: A central fact table is connected to multiple dimension tables. The fact table stores numeric values like sales and quantity, while dimension tables store descriptive data such as product, customer, time, and store. All tables are directly connected without normalization.

Example: Fact Table: Sales (Sales_ID, Amount, Quantity)

Dimension Tables: Product, Customer, Time, Store



Q6. OLAP operations with SQL examples

Ans:

Roll-up: Summarizes data to higher level

Example:

SQL

```
SELECT YEAR(date) AS year, SUM(sales)
FROM sales_table
GROUP BY YEAR(date);
```

Drill-down: Breaks data into more detail

Example:

SQL

```
SELECT YEAR(date), MONTH(date), SUM(sales)
```

FROM sales_table

GROUP BY YEAR(date), MONTH(date);

Slice: Filters data on one dimension

Example:

SQL

SELECT *

FROM sales_table

WHERE year = 2025;

Dice: Filters on multiple dimensions

Example:

SQL

SELECT *

FROM sales_table

WHERE year = 2025 AND product = ‘A’ AND location = ‘Gujarat’;

Q7. Explain the ETL process in data warehousing

Ans:

ETL stands for Extract, Transform, Load. It is used to move data from different sources into a data warehouse.

Extract: Data is collected from sources like databases, files, or APIs.

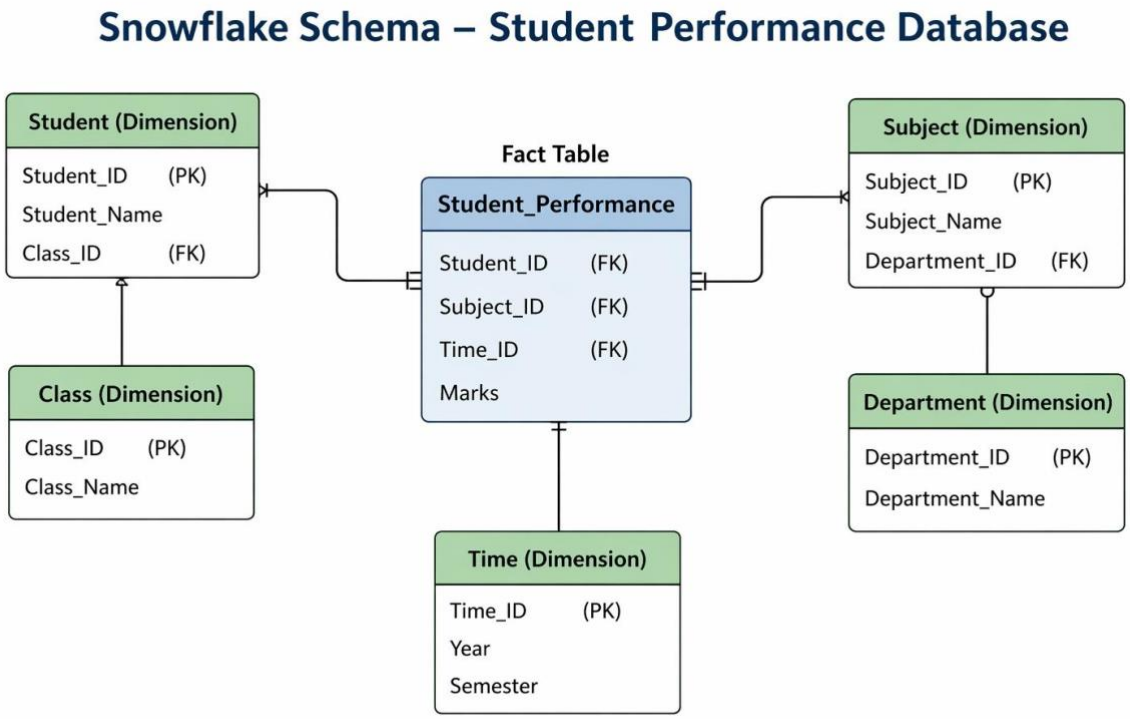
Transform: Data is cleaned, filtered, and converted into a proper format. Errors are removed and data is made consistent.

Load: The transformed data is stored into the data warehouse for analysis.

Example: Data from sales systems is extracted, cleaned (remove duplicates, fix errors), and loaded into a warehouse for reporting.

Q8. Design a Snowflake Schema for student performance database

Ans:



Or

Q9. Apply OLAP operations on a sales dataset and explain results

Ans:

Roll-up

SQL

SELECT Year, SUM(Amount) AS Total_Sales

FROM Sales

GROUP BY Year;

Result: Shows total sales per year.

Drill-down

SQL

SELECT Year, Month, SUM(Amount) AS Total_Sales

FROM Sales

GROUP BY Year, Month;

Result: Shows sales per month within each year.

Slice

SQL

SELECT *

FROM Sales

WHERE Year = 2025;

Result: Shows only 2025 sales data.

Dice

SQL

SELECT *

FROM Sales

WHERE Year = 2025 AND Product = ‘A’ AND Region = ‘Gujarat’;

Result: Shows filtered data for given conditions.

Q10. Define data mining tools and list any three

Data mining tools are software used to analyze large datasets, find patterns, and extract useful information for decision-making.

They provide techniques like classification, clustering, and association rule mining.

These tools help in predicting trends and improving business strategies.

They are widely used in business, healthcare, and education sectors.

Examples: Orange, RapidMiner, KNIME

Or

Q11.List applications of data mining in business, education, and hhealthcare.

Ans:

1. Business

Customer segmentation for targeted marketing

Sales forecasting and trend analysis

Fraud detection in transactions

Recommendation systems (e-commerce)

Inventory and supply chain optimization

2. Education

Predicting student performance and dropout risk

Personalized learning systems

Analysis of student behavior and attendance

Curriculum improvement based on learning patterns

Admission and enrollment trend analysis

3. Healthcare

Disease prediction and early diagnosis

Patient risk analysis (e.g., diabetes, heart disease)

Medical image analysis (X-rays, MRI)

Treatment effectiveness analysis

Hospital resource and patient management optimization

Q12. Explain the features of Orange, RapidMiner, and KKNIME

Ans:

1. Orange

Open-source, visual programming tool for data mining and machine learning

Drag-and-drop interface (no heavy coding required)

Good for beginners and education purposes

Supports data visualization (interactive charts, graphs)

Includes ML algorithms like classification, clustering, regression

Extensible using Python scripts

2. RapidMiner

Powerful data science platform for data mining and predictive analytics

GUI-based workflow design (drag-and-drop process)

Supports automation of data preparation, modeling, and evaluation

Large library of machine learning and statistical models

Handles big data integration and real-time analytics

Strong support for business intelligence and enterprise use

3. KNIME

Open-source data analytics and integration platform

Uses modular workflow system (nodes connected in pipelines)

Supports data preprocessing, mining, and visualization

Easily integrates with Python, R, and Java

Good for big data processing and automation

Strong community and plugin ecosystem

Or

Q13.Discuss the importance of data mining in decision-making.

Ans:

4. Improves decision accuracy – Uses real data patterns for better results
5. Predicts future trends – Helps forecast sales, demand, and behavior
6. Finds hidden patterns – Discovers useful relationships in data
7. Increases efficiency – Optimizes business operations and resources
8. Detects fraud and risks – Identifies unusual activities early
9. Customer-focused decisions – Helps understand user preferences
10. Faster decisions – Reduces time in analyzing large data sets

Q14.Describe how data mining is used in student performance aanalysis.

Ans:

How Data Mining is Used in Student Performance Analysis

1. Performance prediction – Analyzes marks, attendance, and behavior to predict exam results
2. Identifying weak students – Detects students who are likely to fail or need extra support
3. Learning pattern analysis – Studies how students learn and which subjects they struggle with
4. Dropout prediction – Identifies students at risk of leaving the course early

5. Personalized learning – Suggests customized study material based on performance
6. Teacher decision support – Helps educators improve teaching methods using data insights
7. Exam result analysis – Finds trends in subject-wise performance and difficulty levels

Or

Q15. Explain the role of data mining in business iintelligence.

Ans:

Role of Data Mining in Business Intelligence

1. Data analysis for insights – Extracts useful patterns from large business data
2. Better decision-making – Helps managers make informed and accurate decisions
3. Market trend analysis – Identifies customer behavior and market changes
4. Customer segmentation – Groups customers based on buying patterns
5. Sales forecasting – Predicts future sales and demand
6. Fraud detection – Detects unusual transactions and reduces business risk
7. Performance improvement – Optimizes business operations and strategies